

Xinyue (Cindy) Wang

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EDUCATION

Carnegie Mellon University

Major in Computer Science and Information Systems

Pittsburgh, PA

Expected May 2027

- Relevant Coursework: Computer Systems, Imperative Computation, Database Development, Artificial Intelligence, Foundations of Software Engineering, Application Development, Functional Programming

SKILLS

Languages: Python, C, Java, SQL, JavaScript, HTML/CSS, Assembly, R, Ruby

Tools: Git, Docker, GitHub Actions, Flask, Node.js, Redis, React, Rails, Vertabelo, MongoDB, Figma, LaTeX

EXPERIENCE

Researcher - Cyber Defense Automation

Sept 2024 – Present

Carnegie Mellon School of Computer Science, Department of Defense

Pittsburgh, PA

- Constructed a knowledge graph system that models logical dependencies across 183 MITRE D3FEND techniques to automate the generation of response playbooks.
- Engineered Python scripts to structure unstructured technical text into JSON, utilizing cosine similarity to match condition pairs and form logical dependencies.
- Reduced AI hallucinations through iterative validation, achieving 89.7% soundness and 98.9% completeness in the final dataset.

Student Researcher - UC Davis COSMOS

Jul 2023 – Aug 2023

University of California, Davis

- Researched nuclear physics and electro-optics, utilizing calculus to analyze experimental data and model system behavior.

Chief Technology Officer

Aug 2023 – May 2024

Aegis, Inc. Virtual Enterprises International Company

- Elected to direct IT and Marketing departments for a virtual firm competing against 7,000+ international student-led businesses.
- Developed company website that generated \$45,000 in simulated sales in the first 3 months through targeted UX design.
- Presented market research and statistical analysis at Business Plan competitions and Trade Shows, competing against 70 top firms in California.

PROJECTS

AI-Powered Q&A Forum (NodeBB) | Node.js, Python, Docker, Ollama (LLM), GitHub Actions

- Architected a microservice integration using an LLM for automated post-translation, connecting a Node.js frontend to a Python/Flask backend.
- Managed production deployment via Docker, configuring networking and external hosts to enable seamless API communication between distributed containers.

Dynamic Memory Allocator | C, malloc/free, Segregated Free Lists, Heap Utilization

- Built a dynamic memory allocator supporting malloc, free, realloc, and calloc using segregated free lists.
- Optimized block coalescing, splitting, and alignment to improve > 75% heap utilization and > 10000 kops/sec throughput.

Concurrent Web Proxy Server | C, Sockets, HTTP, Pthreads

- Constructed an HTTP/1.0 web proxy that accepts client connections, parses requests, forwards them to end servers, and streams responses back.
- Allowed for concurrency using POSIX threads to handle multiple requests at the same time.

Linux Shell Program | C, Process Control, Signals, I/O Redirection

- Developed a Linux shell supporting job control, background and foreground execution, and I/O redirection, including built-in commands.
- Programmed signal handling and process group management to control running jobs and avoid race conditions.

Multithreaded Filesystem | C, Concurrency, Synchronization, File Systems

- Implemented filesystem features and added thread safe concurrent access using synchronization to prevent conflicting operations.

Instacart Database | SQL, Vertabelo, UML, Relational Modeling, BCNF

- Designed a database from Instacart's user stories by creating a conceptual model, relational model, and physical model.
- Formulated SQL queries for each user story and used functional dependencies, closure, and BCNF normalization to improve the schema.

Cache Simulator | C, Memory Locality, Bit Manipulation, 64-bit Memory Addressing

- Simulated a configurable cache using LRU replacement and write-back and write-allocate policies.
- Reduced cache misses by 75% (from ~1,200 to < 300) for 32x32 matrix through blocking and spatial locality.